

Patient Details

Name	: DEVIKA	Sex / Age	: Female/ 19 Years	Case ID	: 60300129567
Ref By	:	PT. ID	: 2908914 OQG2603250447	Test Name	: ORION (V2) (WES-Whole Exome Sequencing)
Bill. Loc.	: Oncquest Laboratories Ltd.				

Sample Details

Registration Date & Time	: 2026-03-25 16:51:18	Sample Type	: Whole Blood EDTA	Sample Date & Time	: 2026-03-25 16:51:00
Ref ID 1.	: -	Report Date & Time	: 2026-04-19 14:45:21		

Summary of the Clinical History Provided

Salient features: Chronic calcific pancreatitis, intrapancreatic ductal calculi, polypoidal lesion in the distal pancreatic duct
 Clinical suspicion: Pancreatic exocrine insufficiency

Test Results and Interpretation

ON PHENOTYPE BASED ANALYSIS: VARIANT/S DETECTED.

PRSS1 variants are low-depth, low-complexity variants, and hence confirmation via sanger sequencing is recommended.

Summary Of Variants

Gene and Transcript	Exon/Intron Number	Variant Nomenclature	Zygosity	Classification	OMIM Phenotype	Inheritance
	-	p.Ala73Thr [101x /186x]			susceptibility to}	
PRSS1 (NM_002769.5)	Exon 1/Exon 5	c.40C>G p.Leu14Val	Heterozygous	Uncertain significance	Pancreatitis, hereditary	Autosomal dominant
	-	[54x /243x]				
PRSS1 (NM_002769.5)	Exon 2/Exon 5	c.86A>T p.Asn29Ile	Heterozygous	Pathogenic	Pancreatitis, hereditary	Autosomal dominant
	-	[85x /531x]				

Variant Details

CTRC

Variant Nomenclature	c.217G>A (p.Ala73Thr)
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Genomic Nomenclature	chr1:g.157670736>A
Zygosity	Heterozygous

Type of variant	gnomAD frequency	Computational evidences	ClinVar	Null Variant	Downstream Null Variant	Previously reported [reported zygosity]
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Missense Variant	0.0285%	REVEL: 0.972	Yes (Pathogenic/ Likely Pathogenic/ Uncertain Significance)	NA	NA	Yes (heterozygous)
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Experimental studies have shown that this missense change affects CTRC function (Rosendahl J, et al., 2008).

References:

1. Berke G, et al. Risk of chronic pancreatitis in carriers of the c.180C>T (p.Gly60=) CTTC variant: case-control studies and metaanalysis. Pancreatolgy. 2023.
2. Rosendahl J, et al. Chymotrypsin C (CTTC) variants that diminish activity or secretion are associated with chronic pancreatitis. Nat Genet. 2008.

Disease

PANCREATITIS, HEREDITARY [OMIM GENE ID: 601405]

PRSS1-related hereditary pancreatitis (HP) is characterized by episodes of acute pancreatitis (AP) and recurrent acute pancreatitis (RAP: >1 episode of AP), with frequent progression to chronic pancreatitis (CP). Manifests from vague abdominal pain lasting one to three days to severe abdominal pain lasting days to weeks and requiring hospitalization.

PRSS1

Variant Nomenclature	c.40C>G (p.Leu14Val)
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Genomic Nomenclature	chr7:g.142457375C>G
Zygosity	Heterozygous

Type of variant	gnomAD frequency	Computational evidences	ClinVar	Null Variant	Downstream Null Variant	Previously reported [reported zygosity]
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Missense Variant, Splice Region Variant	1.6079%	REVEL: 0.221	Yes (Uncertain significance; Benign; Likely benign)	NA	NA	Yes (Internal Database)
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Recommendations

29. A negative report does not exclude a genetic disorder due to inherent limitations of the assay design.
30. On the background of whole exome sequencing, analysis is limited to provided indications/ requested testing. Hence analysis is limited to a single gene if the same has been requested.
31. As a part of knowledge sharing initiative, all reported variants are submitted in de-identified form in the ClinVar database.
32. Extracted DNA if available after requisitioned testing will be stored as per recommendations. Please note that DNA may degrade over time and this may affect the quality of the stored sample.
33. Collected blood samples are not stored
34. Prenatal samples (AF/CVS/cord blood) are not stored. Extracted DNA if available is stored as acknowledged above.
35. As per PCPNDT fetal gender is not revealed.
36. Maternal cell contamination is recommended for prenatal and POC (product of conception) to ensure accuracy of test results. The same is performed only when requested and on the availability of the maternal sample.
37. Discrepant maternal cell contamination results may rise with use of donor gamete and hence information regarding the same should be provided to the laboratory.
38. Reproductive decision making is not recommended based on variants of uncertain significance.
39. Raw data can be transferred on request and after due consent/assent from the involved patient/ family. Additional charges will be applicable for the same.
40. The test performed by the laboratory with the assumption that the sample belongs to the person herewith mentioned in the requisition form and appropriate consent as well as prenatal counseling including test information has been provided by the referring clinician.
41. Repeat sampling may be required in case of gender discrepancy (unless the same can be attributed to an underlying scientific reason) as well as in rare cases where DNA/data quality prevents further analysis.
42. Reanalysis of data is recommended as deemed necessary by the referring clinician. Additional charges will be applicable for the same.

Technical Notes

Methodology - Massively Parallel Sequencing (Next Generation Sequencing): Genomic DNA from the submitted specimen was enriched for the complete coding regions and splice site junctions of genes listed below using a custom bait- capture system. Paired End Sequencing was performed with 2x100/2x150 chemistry. Reads were assembled and were aligned to reference sequences based on NCBI RefSeq transcripts and human genome build GRCh37/UCSC hg19. Data was filtered and analyzed to identify variants of interest and interpreted in the context of a single most damaging, clinically relevant transcript for the purpose of the report, indicated as a part of variant details. Enrichment and analysis focus on the coding sequence of the indicated transcripts, 5-10bp of flanking intronic sequence, and other specific genomic regions demonstrated to be causative of disease at the time of assay design. Deletion and duplication analysis is performed in cases when indicated but detected variations need to be confirmed by an alternate methodology. Sequence and copy number variants are reported according to the Human Genome Variation Society (HGVS).

Laboratory reporting protocol: The analysis is based on the provided phenotype: relevant HPO terms, curated gene panels and relevant literature is assessed for phenotype based analysis. Variant reporting is limited to exon regions and upto 10 basepairs within exon intron boundaries. Previously reported deep intronic and non coding variants will be included when detected at a depth more than 10X. Variant reporting is performed at a minimum depth of 10X. The gnomAD variant frequency reflects the leftover of hg38 to hg19.

For Mitochondrial Genome Sequencing (if requested): Only phenotype-related Pathogenic and Likely Pathogenic variations reported in the MitoMap database as well as literature are reported. Haplogroups are not analyzed. A list of variants other than the above are available on request. Analyzed genes include: MT-ND1, MT-ND2, MT-ND3, MT-ND4L, MT-ND4, MT-ND5, MT-ND6, MT-CYB, MT-CO1, MT-CO2, MT-CO3, MT-ATP6, MT-ATP8, MT-RNR2, MT-RNR1, MT-RNR2, MT-TA, MT-TB, MT-TN, MT-TD, MT-TC, MT-TE, MT-TQ, MT-TG, MT-TH, MT-TI, MT-TL1, MT-TL2, MT-TK, MT-TM, MT-TE, MT-TP, MT-TS1, MT-TS2, MT-TT, MT-TW, MT-TY, MT-TV.

Tools and Databases employed for analysis: Clinvar, OMIM, HGMD, UCSC genome browser, Uniprot, Ensembl, dbSNP, gnomAD, ExAC, Pubmed, Dgapi, icgc, Kaviar, various bioinformatics analysis, predictive tools and disease specific databases used as available and appropriate. Such tools/databases would be mentioned wherever used.

REVEL: The REVEL score for an individual missense variant can range from 0 to 1, with higher scores reflecting greater likelihood that the variant is disease-causing.

Bioinformatics pipeline version: v16.1.1

Gene Coverage

Gene	Coverage	Gene	Coverage	Gene	Coverage	Gene	Coverage
CASR	100%	CEL	96.2%	CFTR	100%	CLDN2	100%
CEP1	100%	CFTRC	100%	GGT1	100%	PBX1	100%
DNILP	100%	PRSS1	100%	PRSS8	100%	SRDS	100%
SPINK1	100%	UBR1	98%				

QC Metrics

Total aligned reads		99.95 %
Total reads	139.19 (M)	
Total data generated	20.58 (Gb)	
Total reads which passed mapping quality cut-off	10.70 (Gb)	

Reviewed By



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